

GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Plan de Studiu — GCP Associate Cloud Engineer

GCP Associate Cloud Engineer

Study Notes Study Notes April 2026

Pentru cineva cu experienta AWS/GCP de baza. Durata recomandata: **4-6 saptamani** (1-2 ore/zi)

Principii de Invatare

Cum sa citesti materialele

1. **Nu citi pasiv** — Dupa fiecare sectiune, inchide fisierul si incearca sa raspunzi: "Ce am invatat? Ce serviciu as folosi pentru X?"
2. **Concentreaza-te pe DIFERENTE** — Ai experienta AWS. Creierul tau va retine cel mai bine lucrurile care sunt *diferite* fata de ce stii. Foloseste `gcp-vs-aws.md` ca ancora.
3. **Regula 3-2-1** — Dupa fiecare sesiune de studiu:
 - **3** lucruri noi invatate
 - **2** lucruri pe care vrei sa le aprofundezi
 - **1** intrebare la care nu stii raspunsul
4. **Invata in spirala, nu liniar** — Treci prin tot materialul o data (surface pass), apoi revino si aprofundeaza (deep pass), apoi testeaza-te (test pass).

Cum sa intelegi (nu sa memorezi)

- **Intreaba-te "De ce?"** — Nu memora "Cloud Run scaleaza la zero". Intelege DE CE: e serverless, platesti per request, deci daca nu sunt requesturi, nu ai nevoie de instante.
- **Compara mereu** — "De ce Spanner si nu Cloud SQL?" → Spanner e global + horizontal scaling. Cloud SQL e regional + vertical scaling. Pretul reflecta asta.
- **Gandeste in scenarii** — "Sunt CTO la un startup. Am o aplicatie web cu trafic imprevizibil. Ce aleg?" → Cloud Run (scale to zero, cost minim, no ops).

- **Deseneaza** — Deschide un caiet si deseneaza arhitecturi. VPC-uri cu subnete, LB-uri cu backend-uri, IAM inheritance. Desenul forteaza intelegerea.

Saptamana 0 — Pregatire (Ziua 1)

Obiectiv: Intelege structura examenului si fa un self-assessment.

Pas	Ce faci	Fisier	Timp
1	Citeste overview-ul examenului	<code>00-exam-overview.md</code>	15 min
2	Parcurge rapid GCP vs AWS	<code>gcp-vs-aws.md</code>	20 min
3	Noteaza ce stii deja si ce e nou	Caiet/notite proprii	10 min

Dupa acest pas ar trebui sa stii: cele 5 sectiuni, procentajele, si unde ai gaps fata de ce stii din AWS.

Saptamana 1 — Fundamente (Sectiunile 1 + 5)

De ce 1 si 5 impreuna? Sunt strans legate — setup + security formeaza baza. Daca nu intelegi IAM si hierarchia, nimic altceva nu are sens.

Zilele 1-2: Resource Hierarchy + IAM

Pas	Ce faci	Fisier	Timp
1	Citeste in profunzime	<code>01-setting-up-cloud-projects-and-accounts.md</code>	40 min
2	Vizualizeaza relatiile	<code>visual-maps/section1-setup.md</code>	20 min
3	Testeaza-te cu flashcards	<code>flashcards/section1-setup.md</code> (Q1-Q20)	15 min

Focus: Org → Folders → Projects → Resources. Policy inheritance (additive!). Diferenta fata de AWS: VPC e global, Project ≈ AWS Account.

Zilele 3-4: IAM + Service Accounts

Pas	Ce faci	Fisier	Timp
1	Citeste IAM in profunzime	18-managing-iam.md	40 min
2	Citeste Service Accounts	19-managing-service-accounts.md	40 min
3	Vizualizeaza	visual-maps/section5-security.md	20 min
4	Flashcards	flashcards/section5-security.md	20 min

Focus: Basic vs Predefined vs Custom roles. Service Account = si principal SI resource. Impersonation > Keys. Workload Identity = IRSA din AWS.

Zilele 5-6: Billing + Cheat Sheet Review

Pas	Ce faci	Fisier	Timp
1	Citeste billing	02-managing-billing-configuration.md	30 min
2	Review rapid cheat sheets	cheatsheets/section1-setup.md + cheatsheets/section5-security.md	20 min
3	Flashcards complete sectiunile 1+5	flashcards/section1-setup.md + flashcards/section5-security.md	20 min

Focus: Budgets NU opresc cheltuielile (trebuie Pub/Sub + Cloud Function). Billing export → BigQuery. SUDs sunt automate (nu exista in AWS!).

Ziua 7: Recapitulare Saptamana 1

- Parcurge cheat sheets section 1 + 5 fara sa te uiti la raspunsuri
- Fa scenariile relevante din [scenarios/practice-scenarios.md](#) (16-19, 24, 28)
- Noteaza ce inca nu intelegi → revii saptamana urmatoare

Saptamana 2 — Planning (Sectionea 2)

Aceasta e saptamana deciziilor: "Ce serviciu folosesc pentru X?"

Zilele 1-2: Compute

Pas	Ce faci	Fisier	Timp
1	Citeste compute planning	03-planning-compute-resources.md	45 min
2	Studiaza decision flowcharts	visual-maps/section2-planning.md (compute)	20 min
3	Flashcards compute	flashcards/section2-planning.md (Q1-Q15)	15 min

Focus: Flowchart-ul de decizie: CE vs GKE vs Cloud Run vs Functions. Cand Spot VMs. Custom machine types. GKE Standard vs Autopilot.

Zilele 3-4: Databases + Storage

Pas	Ce faci	Fisier	Timp
1	Citeste data storage	04-planning-data-storage-options.md	45 min
2	Studiaza database decision tree	visual-maps/section2-planning.md (databases)	20 min
3	Flashcards databases	flashcards/section2-planning.md (Q16-Q30)	15 min

Focus: OLTP vs OLAP. Cloud SQL (< 64TB, regional) vs Spanner (global, unlimited). Firestore modes (Native vs Datastore — NU se poate schimba!). Storage classes — TOATE au ms latency (spre deosebire de Glacier!).

Zilele 5-6: Networking + Review

Pas	Ce faci	Fisier	Temp
1	Citeste networking	<code>05-planning-network-resources.md</code>	35 min
2	Studiază LB flowchart	<code>visual-maps/section2-planning.md</code> (networking)	15 min
3	Cheat sheet complet	<code>cheatsheets/section2-planning.md</code>	15 min
4	Flashcards restul	<code>flashcards/section2-planning.md</code> (Q31-Q40)	10 min

Focus: 6 tipuri de load balancer (memorizează flowchart-ul!). Premium vs Standard tier. VPC e GLOBAL in GCP (mare diferență față de AWS!).

Ziua 7: Recapitulare Saptamana 2

- Scenarii din `scenarios/practice-scenarios.md` (1-12)
- Exercițiu: desenează pe hartie flowchart-ul de decizie compute + database din memorie

Saptamana 3 — Deploying (Secțiunea 3, partea 1)

Cea mai grea secțiune (25% din examen). Aici trebuie să știi COMENZI.

Zilele 1-2: Compute Engine + MIGs

Pas	Ce faci	Fisier	Temp
1	Citeste deploying CE	<code>06-deploying-compute-engine.md</code>	45 min
2	Vizualizează MIG flow	<code>visual-maps/section3-deploying.md</code> (VM flow)	15 min
3	Flashcards	<code>flashcards/section3-deploying.md</code> (Q1-Q15)	15 min

Focus: `gcloud compute instances create` cu parametri cheie. Instance templates sunt IMUTABILE. MIG + autoscaling + autohealing. OS Login > SSH keys. IAP tunnel pentru VM fără IP

extern.

Zilele 3-4: GKE + Cloud Run/Functions

Pas	Ce faci	Fisier	Timp
1	Citeste GKE deploying	07-deploying-gke-resources.md	40 min
2	Citeste Cloud Run/Functions	08-deploying-cloud-run-and-functions.md	35 min
3	Vizualizeaza	visual-maps/section3-deploying.md (GKE + Cloud Run)	15 min
4	Flashcards	flashcards/section3-deploying.md (Q16-Q30)	15 min

Focus: [gcloud container clusters get-credentials](#) (conecteaza kubectl). Private clusters = nodes fara IP extern. Cloud Run revisions + traffic splitting. Cloud Functions Gen 2 > Gen 1. Eventarc.

Zilele 5-6: Data + Networking + IaC

Pas	Ce faci	Fisier	Timp
1	Citeste data solutions	09-deploying-data-solutions.md	30 min
2	Citeste networking	10-deploying-networking-resources.md	30 min
3	Citeste IaC	11-infrastructure-as-code.md	20 min
4	Flashcards	flashcards/section3-deploying.md (Q31-Q50)	15 min

Focus: VPC peering e NON-TRANZITIV. Firewall rules: priority (0=max), tags vs service accounts. Terraform: init → plan → apply. State in GCS, nu local.

Ziua 7: Recapitulare Saptamana 3

- Cheat sheet complet: [cheatsheets/section3-deploying.md](#)
- Scenarii: [scenarios/practice-scenarios.md](#) (13-15, 25-27)

Saptamana 4 — Operations (Sectiunea 4)

Zilele 1-2: Compute + GKE Management

Pas	Ce faci	Fisier	Temp
1	Citeste managing CE	<code>12-managing-compute-engine.md</code>	35 min
2	Citeste managing GKE	<code>13-managing-gke-resources.md</code>	40 min
3	Vizualizeaza	<code>visual-maps/section4-operations.md</code> (snapshots, autoscaling)	15 min
4	Flashcards	<code>flashcards/section4-operations.md</code> (Q1-Q20)	15 min

Focus: Snapshots = incremental, backup. Images = golden image, templates. HPA (pods) vs VPA (resources) vs Cluster Autoscaler (nodes). StatefulSets pentru databases.

Zilele 3-4: Cloud Run + Storage + Networking

Pas	Ce faci	Fisier	Temp
1	Citeste managing Cloud Run	<code>14-managing-cloud-run.md</code>	25 min
2	Citeste managing storage	<code>15-managing-storage-and-databases.md</code>	35 min
3	Citeste managing networking	<code>16-managing-networking-resources.md</code>	25 min
4	Flashcards	<code>flashcards/section4-operations.md</code> (Q21-Q40)	15 min

Focus: Cloud Run: `--no-traffic` + tags pentru canary. Lifecycle policies pentru cost optimization. Subnet expansion = one-way (nu poti micsora). Cloud NAT = outbound only.

Zilele 5-6: Monitoring + Logging

Pas	Ce faci	Fisier	Timp
1	Citeste monitoring/logging	17-monitoring-and-logging.md	45 min
2	Vizualizeaza log routing	visual-maps/section4-operations.md (logging)	15 min
3	Flashcards	flashcards/section4-operations.md (Q41-Q50)	10 min
4	Cheat sheet	cheatsheets/section4-operations.md	15 min

Focus: Admin Activity logs = mereu active, gratuite. Data Access = optional, cu cost. Log sinks + writer identity (trebuie permisiuni pe destinatie!). Ops Agent = inlocuieste agentii vechi.

[_Required](#) bucket = 400 zile, nu se poate modifica.

Ziua 7: Recapitulare Saptamana 4

- Scenarii: [scenarios/practice-scenarios.md](#) (20-22)
- Parcurge toate cheat sheets (1-5) rapid

Saptamana 5 — Consolidare + Testare

Zilele 1-2: Full Review

Pas	Ce faci	Fisier	Timp
1	Parcurge TOATE cheat sheets	cheatsheets/section1-5	30 min
2	Parcurge TOATE visual maps	visual-maps/section1-5	45 min
3	Noteaza conceptele care inca te incurca	Caiet propriu	15 min

Zilele 3-4: Flashcards Marathon

Pas	Ce faci	Fisier	Timp
1	Toate flashcards, sectiunile 1-5	<code>flashcards/*</code> (~200 intrebari)	60 min/zi
2	Marcheaza intrebarile gresite	Caiet propriu	—
3	Reciteste materialul doar pentru greseli	Fisierele relevante	30 min

Zilele 5-6: Scenarii Complete

Pas	Ce faci	Fisier	Timp
1	Fa TOATE 28 scenariile	<code>scenarios/practice-scenarios.md</code>	60 min/zi
2	Pentru fiecare raspuns gresit, citeste explicatia	—	—
3	Revizuieste sectiunile slabe	Fisierele relevante	30 min

Ziua 7: Practice Exam

- Fa examenul gratuit de practica de la Google: cloud.google.com/certification/practice-exam/cloud-engineer
- Noteaza scorul si sectiunile slabe

Saptamana 6 — Ultimele Zile Inainte de Examen

Zilele 1-3: Focus pe Puncte Slabe

- Reciteste DOAR sectiunile unde ai gresit la practice exam
- Repeta flashcards-urile doar pentru acele sectiuni

Zilele 4-5: Speed Review

Pas	Ce faci	Timp
1	Toate cheat sheets (5 fisiere)	25 min
2	<code>gcp-vs-aws.md</code> — diferentele cheie	10 min
3	Scenariile gresite anterior	20 min

Ziua 6: Ziua Dinaintea Examenului

- Parcurge `cheatsheets/` o ultima data — 25 minute
- NU invata nimic nou
- Odihna, somn bun

Ziua 7: EXAMEN

- Regula: ~2 minute per intrebare
- Flag intrebarile dificile si revino la sfarsit
- Citeste TOATE optiunile inainte sa raspunzi
- Elimina raspunsurile evident gresite mai intai

Ordinea de Citire Recomandata (Rezumat)

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SAPTAMANA 1: 01 → 18 → 19 → 02      (Fundamente: Hierarchy + IAM + Billing)
SAPTAMANA 2: 03 → 04 → 05              (Planning: Ce serviciu pentru ce?)
SAPTAMANA 3: 06 → 07 → 08 → 09 → 10 → 11 (Deploying: Cum fac deploy?)
SAPTAMANA 4: 12 → 13 → 14 → 15 → 16 → 17 (Operations: Cum administrez?)
SAPTAMANA 5: Cheat sheets + Flashcards + Scenarii (Consolidare)
SAPTAMANA 6: Puncte slabe + Speed review  (Final prep)

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Sfaturi de Examen (Exam Day Tips)

1. **Citeste intrebarea de 2 ori** — Multe intrebari au cuvinte cheie: "MOST cost-effective", "LEAST operational overhead", "MINIMUM permissions"
2. **Elimina raspunsurile gresite** — De obicei poti elimina 2 din 4 imediat

3. **"Lift and shift" = Compute Engine** — Daca intrebarea zice "migrate as-is", raspunsul e aproape mereu CE
 4. **"Minimize operational overhead" = Serverless** — Cloud Run > GKE > Compute Engine
 5. **"Global + strongly consistent" = Spanner** — Singurul serviciu cu aceasta combinatie
 6. **"Least privilege" = Predefined role** sau **Custom role** — Niciodata basic roles
 7. **"Without service account keys" = Workload Identity** sau **Impersonation**
 8. **Bugete NU opresc cheltuielile** — Trebuie Pub/Sub + Cloud Function
 9. **VPC Peering e non-tranzitiv** — $A \leftrightarrow B$ si $B \leftrightarrow C$ NU inseamna $A \leftrightarrow C$
 10. **Audit logs: Admin Activity = mereu activ, gratuit**
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Checklist Final — Esti Pregatit?

- ☐ Pot desena resource hierarchy din memorie (Org → Folders → Projects → Resources)
 - ☐ Pot alege serviciul de compute corect pentru orice scenariu
 - ☐ Pot alege baza de date corecta pentru orice scenariu
 - ☐ Stiu cele 6 tipuri de load balancer si cand le folosesc
 - ☐ Inteleg IAM: basic vs predefined vs custom roles
 - ☐ Inteleg Service Accounts: creare, atribuire, impersonation, Workload Identity
 - ☐ Stiu comenzile gcloud esentiale pentru fiecare serviciu
 - ☐ Inteleg VPC: subnets, firewall rules, peering, VPN, NAT
 - ☐ Inteleg monitoring: alerts, log sinks, audit logs, Ops Agent
 - ☐ Pot explica diferentele cheie GCP vs AWS
 - ☐ Am facut practice exam-ul Google si am obtinut 80%+
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